RAJALAKSHMI ENGINEERING COLLEGE

25 years OF ACADEMIC EXCELLENCE

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

FINPILOT AI

An Agentic AI-Based Autonomous Personal Financial Planning and Decision Support System

AD23731 - FOUNDATIONS OF AGENTIC AI

Team Members

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PROBLEM STATEMENT & MOTIVATION

Problem Statement

Most personal finance applications mainly focus on expense tracking, transaction categorization, and static budgeting.

However, users still need to manually:

- Interpret their financial data
- Decide how much to save
- Adjust spending habits
- Plan for financial goals
- Track whether their plan is working

These systems provide information, but they do not actively reason, plan, monitor, or adapt to the user's changing financial situation.

Motivation

- FinPilot AI is designed to transform personal finance management from a passive tracking system into an intelligent decision-support system.
- It uses multiple AI agents to:

Analyze → Reason → Plan → Recommend → Monitor → Adapt

- The system aims to provide personalized financial guidance based on the user's income, expenses, savings behaviour, and financial goals.

"From passive financial tracking to adaptive financial planning."

PROJECT OBJECTIVES

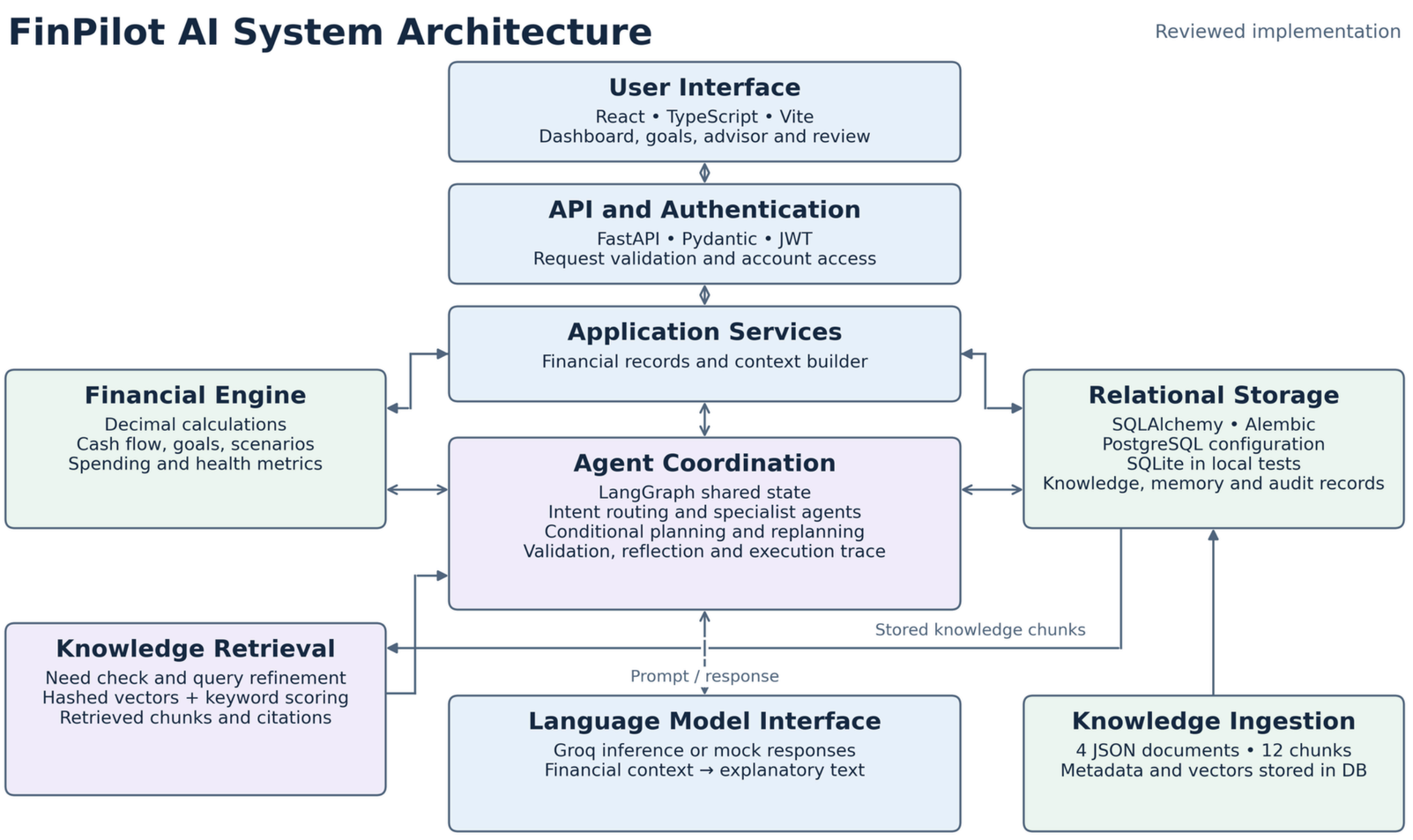
The main objective of FinPilot AI is to develop an agentic personal finance system that can understand a user's financial situation and provide adaptive decision support.

Key Objectives

- Analyze user income, expenses, savings, and spending patterns.
- Identify financial issues and areas of overspending.
- Generate personalized monthly budgets.
- Support goal-based financial planning.
- Provide context-aware financial recommendations.
- Monitor progress toward financial goals.
- Adapt recommendations when income or expenses change.
- Coordinate multiple specialized AI agents through an orchestration workflow.

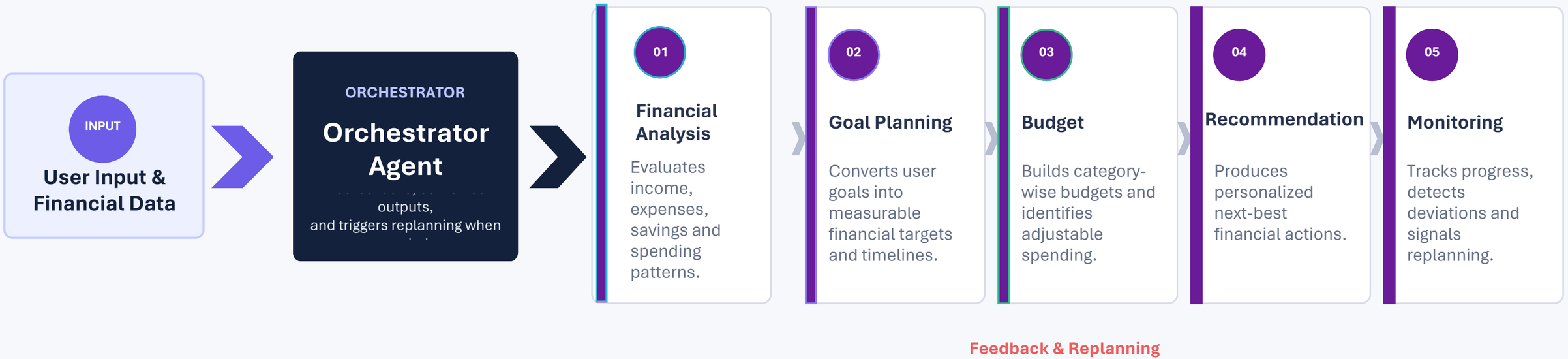
Observe → Analyze → Plan → Recommend → Monitor → Adapt

OVERALL SYSTEM ARCHITECTURE



FinPilot AI follows a layered architecture that connects the user interface, financial data, AI agents, and reasoning components into a single autonomous workflow.

AGENT ORCHESTRATION WORKFLOW



How it works

Each agent completes a focused task. Its output becomes context for the next agent, while the Monitoring Agent can send control back to the Orchestrator when the user's financial situation changes.

AGENTIC CYCLE

OBSERVE → REASON → PLAN → ACT → MONITOR → ADAPT

AI AGENTS & RESPONSIBILITIES

Specialized agents collaborate to analyze, plan, recommend, and adapt to the user's financial situation.

01 Orchestrator Agent

Coordinates the workflow, routes tasks between agents, and combines outputs into one financial plan.

02 Financial Analysis Agent

Analyzes income, expenses, savings, and spending patterns to understand the user's financial condition.

03 Goal Planning Agent

Transforms user goals into measurable targets, timelines, and actionable planning steps.

04 Budget Agent

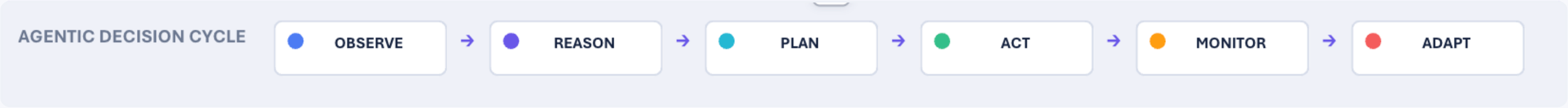
Creates and adjusts category-wise budgets based on income, expenses, and financial priorities.

05 Recommendation Agent

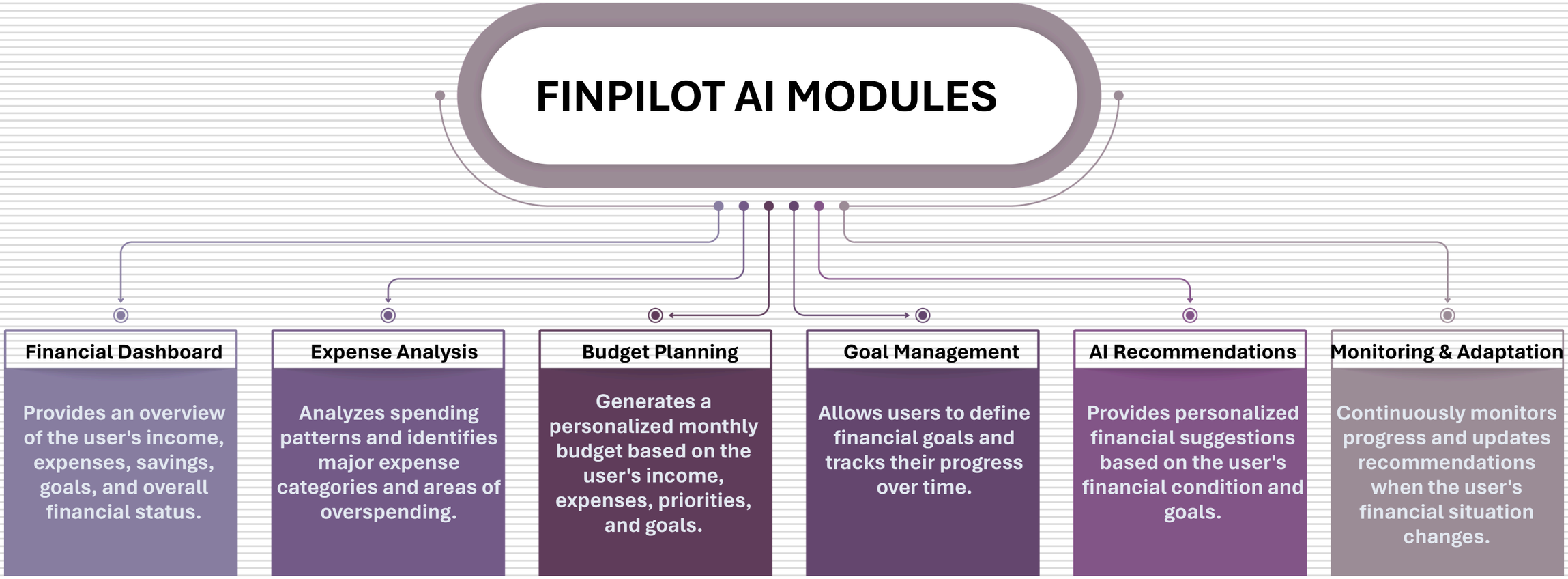
Generates context-aware financial suggestions using the outputs of the other specialized agents.

06 Monitoring Agent

Tracks progress, detects deviations, and triggers feedback or replanning when conditions change.



MAJOR SYSTEM MODULES



TECHNOLOGY STACK

FinPilot AI is implemented using a modular architecture that combines the user interface, backend services, AI reasoning, and multi-agent coordination.

Technology Stack

- Frontend: User interface for financial inputs, dashboards, goals, and recommendations
- Backend: Handles application logic, financial processing, and API communication
- AI Layer: Large Language Model based reasoning and prompt engineering
- Agentic Layer: Multi-agent architecture with orchestrated task execution
- Database: Stores user financial data, goals, budgets, and monitoring history
- Development Tools: Git, GitHub, and VS Code

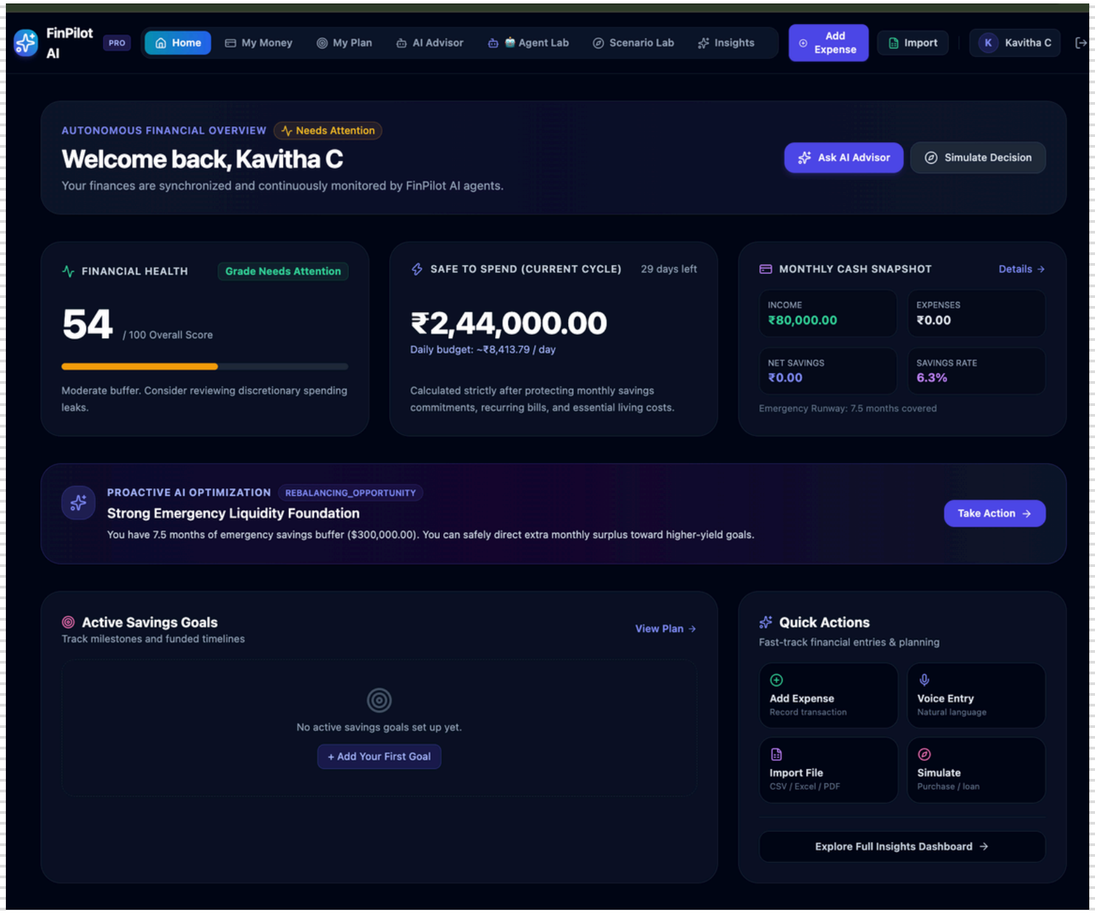
Implementation Flow

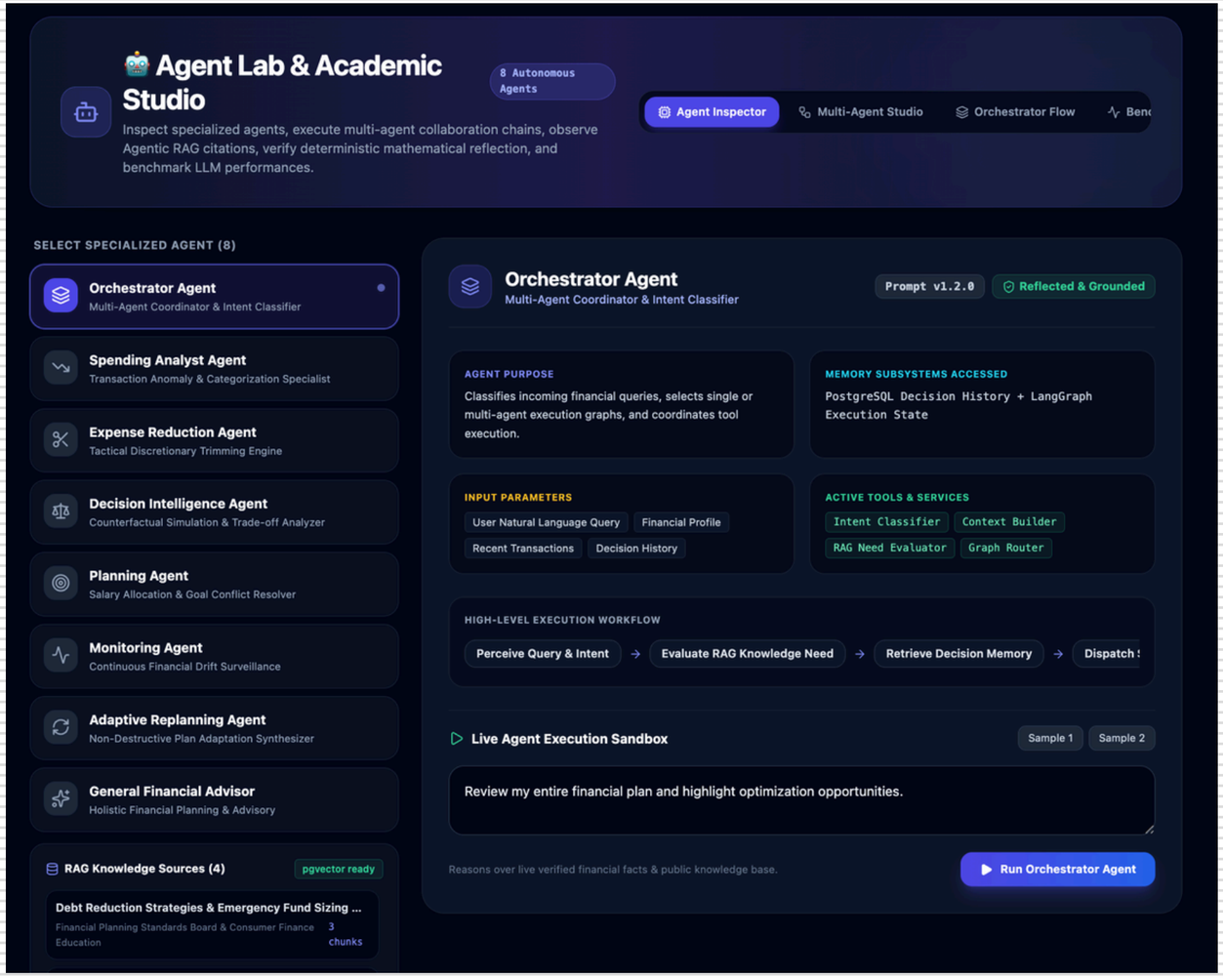
User Input → Backend Processing → Agent Orchestration → AI Reasoning → Financial Recommendation → Monitoring & Feedback

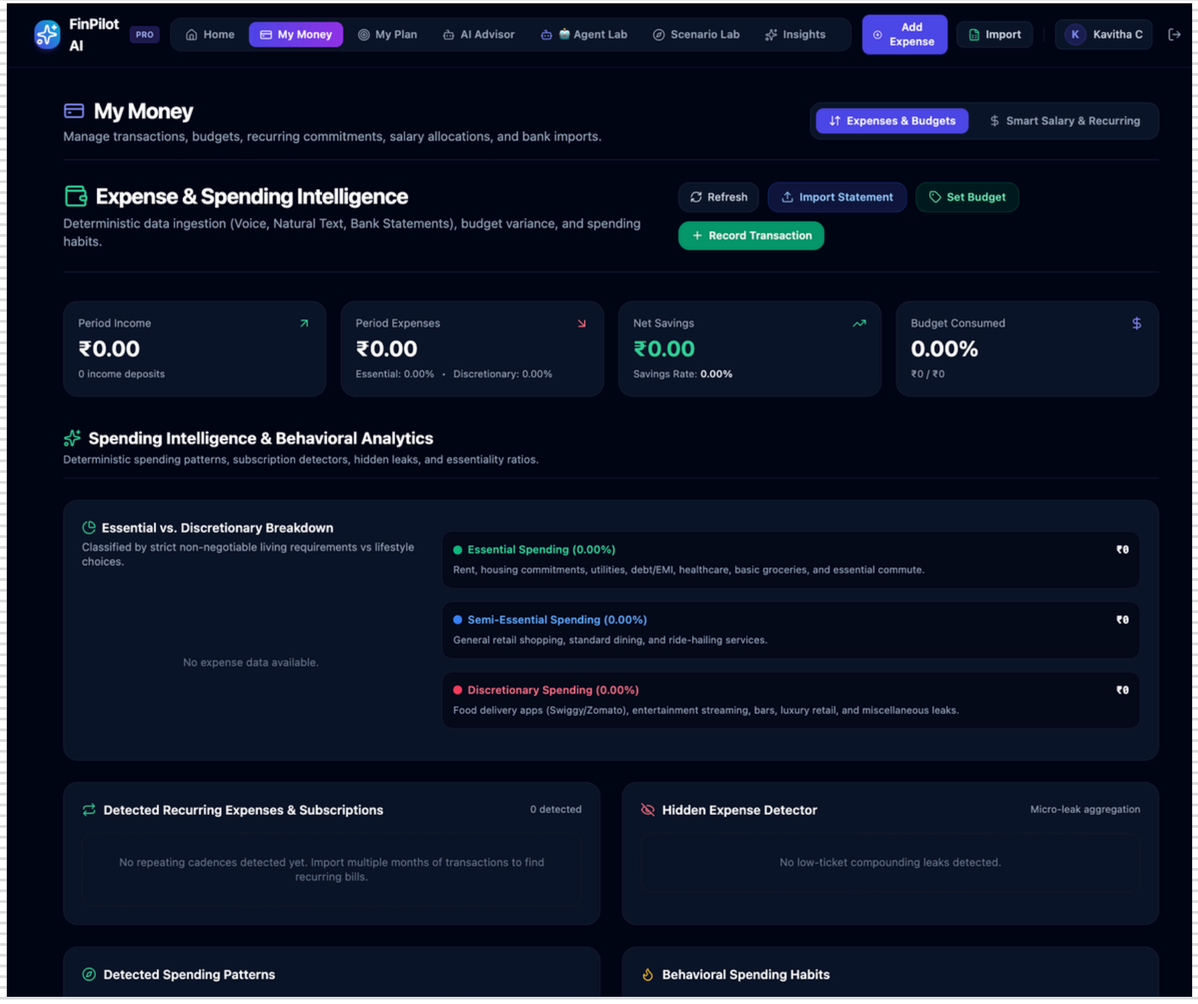
RESULTS & APPLICATION OUTPUT

Key Outcomes

- Financial data is analyzed to understand the user's current financial condition.
- Spending patterns and important expense categories are identified.
- Personalized budgets are generated based on income and expenses.
- Financial goals are converted into actionable plans.
- Multiple agents coordinate to generate context-aware recommendations.
- Progress is monitored and recommendations can be updated when financial conditions change.




Agent Lab & Academic Studio
8 Autonomous Agents
Inspect specialized agents, execute multi-agent collaboration chains, observe Agentic RAG citations, verify deterministic mathematical reflection, and benchmark LLM performances.
Agent Inspector
Multi-Agent Studio
Orchestrator Flow
SELECT SPECIALIZED AGENT (8)
Orchestrator Agent
Multi-Agent Coordinator & Intent Classifier
Spending Analyst Agent
Transaction Anomaly & Categorization Specialist
Expense Reduction Agent
Tactical Discretionary Trimming Engine
Decision Intelligence Agent
Counterfactual Simulation & Trade-off Analyzer
Planning Agent
Salary Allocation & Goal Conflict Resolver
Monitoring Agent
Continuous Financial Drift Surveillance
Adaptive Replanning Agent
Non-Destructive Plan Adaptation Synthesizer
General Financial Advisor
Holistic Financial Planning & Advisory
RAG Knowledge Sources (4)
pgvector ready
Debt Reduction Strategies & Emergency Fund Sizing ...
Financial Planning Standards Board & Consumer Finance Education
3 chunks
Orchestrator Agent
Multi-Agent Coordinator & Intent Classifier
Prompt v1.2.0
Reflected & Grounded
AGENT PURPOSE
Classifies incoming financial queries, selects single or multi-agent execution graphs, and coordinates tool execution.
MEMORY SUBSYSTEMS ACCESSED
PostgreSQL Decision History + LangGraph Execution State
INPUT PARAMETERS
User Natural Language Query
Financial Profile
Recent Transactions
Decision History
ACTIVE TOOLS & SERVICES
Intent Classifier
Context Builder
RAG Need Evaluator
Graph Router
HIGH-LEVEL EXECUTION WORKFLOW
Perceive Query & Intent
Evaluate RAG Knowledge Need
Retrieve Decision Memory
Live Agent Execution Sandbox
Sample 1
Sample 2
Review my entire financial plan and highlight optimization opportunities.
Reasons over live verified financial facts & public knowledge base.
Run Orchestrator Agent


FinPilot AI
PRO
Home
My Money
My Plan
AI Advisor
Agent Lab
Scenario Lab
Insights
Add Expense
Import
Kavitha C
My Money
Manage transactions, budgets, recurring commitments, salary allocations, and bank imports.
Expenses & Budgets
Smart Salary & Recurring
Expense & Spending Intelligence
Deterministic data ingestion (Voice, Natural Text, Bank Statements), budget variance, and spending habits.
Refresh
Import Statement
Set Budget
Record Transaction
Period Income
₹0.00
0 income deposits
Period Expenses
₹0.00
Essential: 0.00% · Discretionary: 0.00%
Net Savings
₹0.00
Savings Rate: 0.00%
Budget Consumed
0.00%
₹0 / ₹0
Spending Intelligence & Behavioral Analytics
Deterministic spending patterns, subscription detectors, hidden leaks, and essentiality ratios.
Essential vs. Discretionary Breakdown
Classified by strict non-negotiable living requirements vs lifestyle choices.
No expense data available.
Essential Spending (0.00%)
₹0
Rent, housing commitments, utilities, debt/EMI, healthcare, basic groceries, and essential commute.
Semi-Essential Spending (0.00%)
₹0
General retail shopping, standard dining, and ride-hailing services.
Discretionary Spending (0.00%)
₹0
Food delivery apps (Swiggy/Zomato), entertainment streaming, bars, luxury retail, and miscellaneous leaks.
Detected Recurring Expenses & Subscriptions
0 detected
No repeating cadences detected yet. Import multiple months of transactions to find recurring bills.
Hidden Expense Detector
Micro-leak aggregation
No low-ticket compounding leaks detected.
Detected Spending Patterns
Behavioral Spending Habits

THANK YOU